

```
1
2  /*
3  A Java Program for Matrix Multiplication of two 2D
4  matrices.
5  The number of rows and columns will be provided by
6  a user.
7  */
8  import java.util.Scanner;
9
10 public class TwoDimensionalMatrixMultiplication {
11     public static void main(String[] args) {
12         int n_rows_1, n_cols_1, n_rows_2, n_cols_2;
13         Scanner scn = new Scanner(System.in);
14         System.out.println("Enter the row and col of matrix 01:");
15         n_rows_1 = scn.nextInt();
16         n_cols_1 = scn.nextInt();
17         System.out.println("Enter the row and col of matrix 02:");
18         n_rows_2 = scn.nextInt();
19         n_cols_2 = scn.nextInt();
20         if (n_cols_1 == n_rows_2) {
21             double matrix01[][] = new double[n_rows_1][n_cols_1];
22             double matrix02[][] = new double[n_rows_2][n_cols_2];
23             double matrix_result[][] = new double[n_rows_1][n_cols_2];
24
25             System.out.println("Enter the values of matrix 01");
26             for (int i = 0; i < matrix01.length; i++) {
27                 for (int j = 0; j < matrix01[i].length; j++) {
28                     System.out.println("There the value of ("
29                         + i + ", " + j + ") location.");
30                     matrix01[i][j] = scn.nextDouble();
31                 }
32             }
33             System.out.println("Enter the values of matrix 02");
34             for (int i = 0; i < matrix02.length; i++) {
35                 for (int j = 0; j < matrix02[i].length; j++) {
36                     System.out.println("There the value of ("
37                         + i + ", " + j + ") location.");
38                     matrix02[i][j] = scn.nextDouble();
39                 }
40             }
41             for (int k = 0; k < n_cols_1; k++) {
42                 for (int i = 0; i < n_rows_1; i++) {
43                     for (int j = 0; j < n_cols_2; j++) {
44                         matrix_result[i][j] += matrix01[i][k] * matrix02[k][j];
45                     }
46                 }
47             }
48             System.out.println("The result after multiplication:");
49             for (int i = 0; i < matrix_result.length; i++) {
50                 for (int j = 0; j < matrix_result[i].length; j++) {
51                     System.out.print(matrix_result[i][j] + " ");
52                 }
53                 System.out.println();
54             }
55         } else {
56             System.err.println("There is a mismatch between col of matrix 01"
57                 + " and row of matrix 02.");
58         }
59         scn.close();
60     }
61 }
```